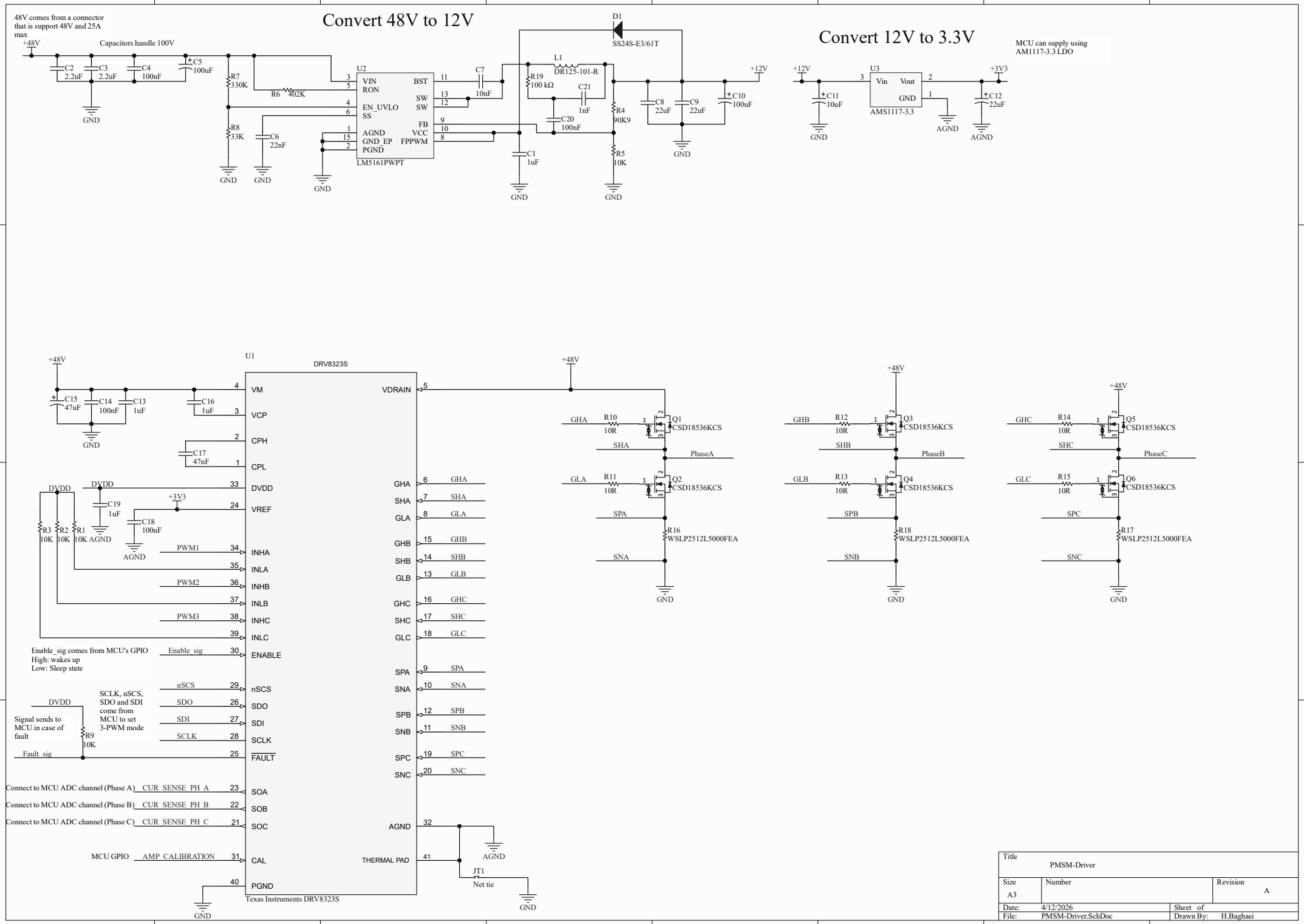
Convert 48V to 12V
48V comes from a connector that is support 48V and 25A max
Capacitors handle 100V
+48V
C2 2.2uF
C3 2.2uF
C4 100nF
C5 100uF
R7 330K
R6 402K
R8 33K
C6 22nF
GND
U2
VIN
RON
EN_UVLO
SS
AGND
GND_EP
PGND
BST
SW
SW
FB
VCC
FPPWM
LM5161PWPT
C7 10nF
C1 1uF
D1
SS24S-E3/61T
L1
DR125-101-R
R19 100 kΩ
C21 1nF
C20 100nF
R4 90K9
R5 10K
C8 22nF
C9 22nF
C10 100uF
+12V
Convert 12V to 3.3V
MCU can supply using AM1117-3.3 LDO
U3
Vin
Vout
GND
AMS1117-3.3
C11 10uF
C12 22uF
+3V3
AGND
U1
DRV8323S
VM
VCP
CPH
CPL
DVDD
VREF
INHA
INLA
INHB
INLB
INHC
INLC
ENABLE
nSCS
SDO
SDI
SCLK
FAULT
SOA
SOB
SOC
CAL
PGND
VDRAIN
GHA
SHA
GLA
GHB
SHB
GLB
GHC
SHC
GLC
SPA
SNA
SPB
SNB
SPC
SNC
AGND
THERMAL PAD
C15 47uF
C14 100nF
C13 1uF
C16 1uF
C17 47nF
C19 1uF
C18 100nF
R3 10K
R2 10K
R1 10K
PWM1
PWM2
PWM3
Enable_sig
Enable_sig comes from MCU's GPIO
High: wakes up
Low: Sleep state
SCLK, nSCS, SDO and SDI come from MCU to set 3-PWM mode
Signal sends to MCU in case of fault
R9 10K
Fault_sig
Connect to MCU ADC channel (Phase A) CUR_SENSE_PH_A
Connect to MCU ADC channel (Phase B) CUR_SENSE_PH_B
Connect to MCU ADC channel (Phase C) CUR_SENSE_PH_C
MCU GPIO AMP_CALIBRATION
Texas Instruments DRV8323S
JT1
Net tie
R10 10R
R11 10R
Q1 CSD18536KCS
Q2 CSD18536KCS
R16 WSLP2512L5000FEA
PhaseA
R12 10R
R13 10R
Q3 CSD18536KCS
Q4 CSD18536KCS
R18 WSLP2512L5000FEA
PhaseB
R14 10R
R15 10R
Q5 CSD18536KCS
Q6 CSD18536KCS
R17 WSLP2512L5000FEA
PhaseC
Title
PMSM-Driver
Size
A3
Number
Revision
A
Date: 4/12/2026
Sheet of
File: PMSM-Driver.SchDoc
Drawn By: H.Baghaei